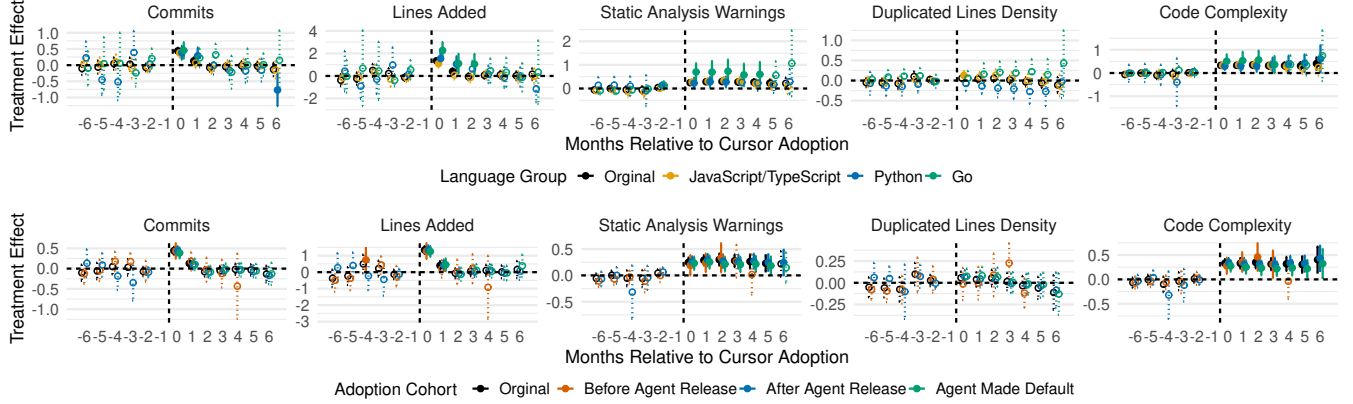


Table 7: Summary statistics for all alternative dataset settings explored in this study. The results from Rust repositories are dropped in Figure 7 because of insufficient observations, causing insignificant results and extremely large confidence intervals.

Setting	# Treatment Repos	# Control Repos	# Total Observations	# Post-Treatment Observations
Main Settings				
Original	801	1,172	21,699	4,481
High Contributor Adoption	379	535	8,440	1,919
Cursor Configuration Changes	801	1,127	18,319	1,569
Active Months (>0 commits)	801	1,172	16,075	3,934
Very Active (≥ 10 commits)	709	814	10,134	2,721
Cursor and Others	382	583	11,322	2,440
Only Cursor	419	602	10,232	2,041
Adoption Cohort Settings				
Before Agent Release	166	286	4,793	1,262
After Agent Release	635	769	15,508	3,219
Agent Made Default	461	570	11,748	2,199
Programming Language Settings				
JavaScript/TypeScript	411	422	8,870	2,279
Python	121	127	2,461	582
Go	35	57	1,147	191
Rust	21	41	771	132

**Figure 7: The estimated “horizon-average” treatment effects in alternative dataset settings with all five outcome variables. Row 1: Robustness check across programming language groups with statistically sufficient observations in our dataset (JavaScript/TypeScript, Python, Go), showing that our findings are qualitatively consistent across programming languages. Row 2: Robustness checks across Cursor adoption cohorts where different Cursor features were available, showing no qualitative difference across cohorts before/after agent release and agent made default in the Composer feature.**

Another limitation of our study is that our identification based on Cursor rule files cannot clearly distinguish developers who use the older Cursor Composer feature (with autonomous file editing but without agentic capabilities) from those who use the later Agentic features. The reality is probably that most repositories in our dataset have a mix of developers using different AI autonomy levels, with most switching to full-fledged agents after it became the default in February 2025. As another robustness check, we check whether the availability of the agentic feature impacts our main results through different adoption cohorts, *Before Agent Release*, for the adoption cohorts before November 2024, *After Agent Release* for the adoption cohorts after November 2024, and *Agent Made Default* for

the adoption cohorts after February 2025. Note that clear separation of feature availability is impossible in our longitudinal study setting, as it is very likely that early Cursor adopters will also switch to using agents later during our observation period. The results in Figure 7, Row 2 show no qualitative difference across the three adoption cohorts for the main findings, except that the estimated effects on static analysis warnings and code complexity are slightly weaker. As with the programming language case, the interpretation of this difference is challenging. The effect may be because of an increase in model capabilities, merely selection bias, or because most repositories eventually switch to agents.